Chapter 14

Multiple sclerosis

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Multiple sclerosis (MS) is a progressive demyelinating and neurodegenerative disease of the central nervous system (CNS) which gradually results in severe neurological deficits. MS usually presents in early adult life when it has a major impact on family, and on vocational and social life. Since lack of myelin slows down the conduction of the action potentials, MS manifests itself as impaired performance that can have a devastating effect on behaviour. MS usually involves a more or less progressive development of neurological symptoms and behavioural deficits. The exact aetiology and pathogenesis, however, are still unclear despite recent advances in understanding this mysterious disease. The disease is characterized by remissions and relapses, erratic onset and duration of symptoms that flare up acutely. There is a growing body of evidence indicating that relapses in MS are associated with stressful events. Each relapse may involve the same or quite different area of the white matter of the CNS. The remissions are rarely complete and may last for either a long or short period. Since the disease remains poorly understood, there is no effective treatment of the disease process itself. New insights into the pathophysiology, however, are suggesting new strategies for intervention. Among these are several disease-modifying pharmarcological agents, which have been shown to have modest effects in slowing down the progress of the disease and to reduce the severity of relapses. Health professionals working with individuals with MS need to focus on those factors arising from the disease they can prevent or change in order to improve the quality of the individual's life.

EPIDEMIOLOGY AND AETIOLOGY

MS is thought to involve an interplay between genetic and environmental factors, which result in an immunologically mediated inflammatory response within the CNS. Although immunological abnormalities have been consistently reported, the relative role of each component of the immune response in mediating tissue damage, and the extent to which these changes are the cause or consequence of myelin injury, remain to be established

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(Compston 1993a). There is compelling evidence for genetic susceptibility to MS (Compston 1990) while environmental influence upon MS is suggested by variation in disease incidence and prevalence according to geographical areas (Weinshenker 1996).

There is a clear pattern of latitudinal variation in the prevalence of MS. Epidemiological studies indicate that the prevalence of MS is high (>30 per 100 000) among young adults in northern Europe, North America and Australasia, medium (5–30 per 100 000) in southern Europe, southern USA and northern Australia and is less prevalent in the Orient, Africa, South America and India (Weinshenker 1996; Marrie 2004; Cristiano et al 2008). Combined data from epidemiology studies conducted in the USA, UK, Australia and New Zealand show a strong link between latitude and prevalence of MS (Fig. 14.1; Richards et al 2002).

MS affects white races principally, although it has been known to affect black immigrants living in Europe and North America. In Australia and New Zealand, MS is rarely seen in the Aboriginal or Maori races but occurs in the white population. In South Africa, the disease occurs more commonly in English-speaking whites than in Afrikaaners but is not seen in native Africans. There is decreasing incidence of the disease in individuals of northern European ancestry as one approaches the tropics, suggesting that certain environments may be relatively more protective. For example, in Australians of northern European descent and in English-speaking white South Africans, the frequency of MS is only about half that of northern Europe. Age is also a factor in immigrant populations. Dean (1967) showed that the risk of MS is higher for English-speaking South African whites who migrated as adults rather than as children.

Further evidence for genetic susceptibility is provided by epidemiological studies within and between ethnic groups. The results of a survey of individuals with MS in Israel in 1973 showed that the disease was common among immigrants from Europe, and rare among immigrants from Afro-Asian countries (Leibowitz et al 1973). Similarly, immigrants from areas of low risk of MS to an area of high risk (such as the UK) retained the low risk of the area of origin (Dean et al 1976). Studies on familial aggregation showed approximately 15% of individuals with MS have an affected relative. This risk rises to 1 : 50 for offspring and 1 : 20 for siblings of affected persons (Sadovnick et al 1988). MS is 20–40 times more common in first-degree relatives, dropping off rapidly with the degree of relatedness (Kantarci & Wingerchuk 2006). Recently, large international collaborations provided strong evidence for the involvement of at least two cytokine receptor genes in the pathogenesis of MS: the interleukin 7 receptor alpha chain gene (IL7RA) on chromosome 5p13 and the interleukin 2 receptor alpha chain gene (IL2RA) (Svejgaard 2008).

Other potential environmental risk factors in MS include infection, vaccinations, stress, climate and diet. Among these, infection is often touted as a putative causal agent, particularly childhood viral infection (Marrie 2004). Dean (1972) suggested that in countries where the disease is relatively rare, early contact with the causative agent, probably a virus, protects the population by making them immune. On the other hand, there is some evidence that exposure to viral illnesses such as measles, mumps and rubella rather late in childhood may be a factor in those who are at risk of developing the disease. In the past decade a few studies have provided preliminary data indicating that environmental stressors, which are perceived by the person as stressful (e.g., broken relationship, losing job), are associated with subsequent relapses in MS (Mohr et al 2004). The evidence of other risk factors such as climate and diet is less convincing.

It would appear, therefore, that MS does not have a single cause. New episodes of demyelination are more likely to occur following a viral infection or stressful events but no single agent or specific stressor has been implicated, suggesting that in the context of genetic susceptibility, environmental influences and immunological priming, demyelination is a physiological response to many pathogens (Compston 1993a). In summary, the epidemiological evidence implicates environmental factors, including psychosocial stressors, operating against a background of genetic susceptibility or resistance during childhood manifesting as altered immune responsiveness (Compston 1993a).

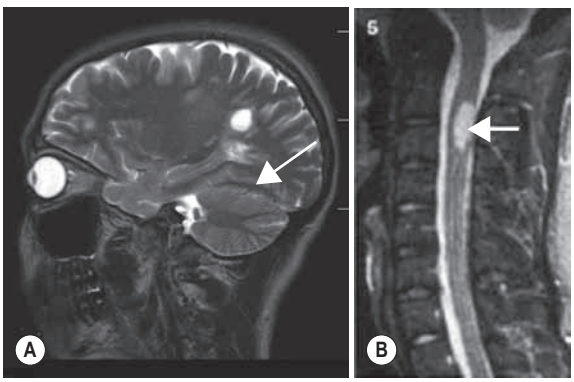

Figure 14.1 An MRI image showing MS lesions in the brain (a) and spinal cord (b).

PATHOPHYSIOLOGY

Demyelination is a disintegration of the myelin sheath caused by an inflammatory and destructive process, the axon being partly or completely denuded. Destruction of the myelin sheath disrupts the normal transmission of

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nerve impulses resulting in neurological signs and symptoms. The axons themselves are preserved initially although some loss of axons may occur, particularly in large chronic plaques.

The hallmark features of MS lesions are perivascular inflammation followed by myelin depletion, oligodendrocyte loss and astroglial proliferation, and these processes are accompanied by limited remyelination and plaque formation. The traditional view suggests that there are four stages of evolution of focal inflammation (Compston 1993b). The initial stage is characterized by the accumulation of inflammatory cells, lymphocytes and monocytes around venules within the CNS. Inflammation is sufficient to cause a functional block in conduction through myelinated axons. Next, there is active destruction of the oligodendrocyte and its myelin sheath as a result of contact with macrophages and microglia. This is followed by depletion of oligodendrocytes in which denuded axons are seen within the lesion. Finally, the lesion heals by scar formation dependent upon astrocytic reactivity, producing hardened patches or plaques from which the disease gets its name.

This view of the mechanism underlying formations of new plaques has recently been challenged. Barnett and colleagues (Barnett & Prineas 2004; Barnett & Sutton 2006), based on recent pathological studies of initial changes in the acute MS lesion, proposed that apoptotic oligodendrocyte death precedes inflammation and demyelination in MS lesions. This means that a MS lesion starts with the death of oligodendrocytes and associated changes within the myelin sheath, which initiates local macrophage scavenger activity with subsequent amplification of inflammatory response. Over months to years the pathology of multiple sclerosis is transformed and the changes which accompany the late phase of the disease suggest that the inflammatory response becomes progressively 'compartmentalized' and therefore largely isolated from systemic influence with time (Barnett & Sutton 2006). However, the causes of oligodendrocyte death are still unknown. In summary, the relative contributions of 'immune-mediated' versus 'neurodegenerative' processes in the disease pathophysiology remain to be answered.

The most common sites of MS lesion are in the grey–white boundary in the cerebrum, the periventricular region, cerebellar white matter, optic nerves and cervical portion of the spinal cord and brainstem, but the disease can involve any part of the CNS (see Fig. 14.1). Until recently MS had been widely considered as a disease of 'white matter' of the CNS. New advances in imaging techniques and post-mortem studies have shown that MS lesions are also present in cortical grey matter, with a particularly high prevalence in progressive forms of the disease (Stadelmann & Bruck 2008). At demyelinated areas, nerve conduction rates are slowed down (Reder & Antel 1983) while conduction along unaffected portions of the axon on either side of the lesion is normal (Smith & McDonald 1999). The conduction block along partially demyelinated axons is the major cause of negative symptoms such as weakness and numbness. It may also explain the fatigue complained of by many patients. These partially demyelinated axons may also discharge spontaneously, accounting for unpleasant distortions of sensation reported by a high percentage of patients (Compston 1993a). Increased temperature sensitivity experienced by many patients after exercise or immersion in hot water may also be explained by the partially demyelinated axons (Compston 1993a).

Remyelination, which involves reinvesting demyelinated axons with new myeline sheaths, does occur in MS and is an important process of 'lesion repair'. Unexpectedly, this process occurs at an early stage of lesion development in both white and grey matter lesions (Stadelmann & Bruck 2008). In contrast, remyelination does not often occur in old lesions where inflammatory activity is minimal. In these older lesions, oligodendrocytes are present but they are not engaged in active remyelination and do not seem capable of producing new myelin (Stadelmann & Bruck 2008). The extent of remyelination in MS, however, is highly variable between patients. Generally, the degree of remyelination of MS lesions is age related and correlates with both number of oligodendrocytes and macrophages in the lesions (Lucchinetti et al 1999). Animal studies (e.g., Zhao et al 2006) suggest that a critical event underlying the failure of remyelination is the impaired ability of phagocytic macrophages to clear myelin debris. This debris contains potent factors that inhibit differentiation of oligodendrocyte precursor cells, which are widely distributed throughout the adult CNS, into oligodendrocytes. Thus, the inflammation response in MS has both destructive (causing demyelination) and beneficial (facilitating remyelination) effects. Understanding of remyelination and why it fails opens up new challenges for future research and could provide development of novel therapeutic strategies (Stadelmann & Bruck 2008).

THE DISEASE PROCESS

Reviews of large databases of MS patients have provided valuable information about its natural history. Clinical signs and symptoms in MS are variable and several patterns can be identified: benign; a relapsing–remitting (RR) course; a secondary progressive (SP) form; and a primary progressive (PP) form. Clinical onset in 80–85% of MS patients is manifested by subclinical neurological symptoms, either multifocal or anatomically discrete, which initially may recover fully. This presentation is often known as a clinically isolated syndrome (CIS). Among patients with CIS, about 20% presented with lesions involving the optic nerve, 45% with long tract signs and

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symptoms, 10% with a brainstem syndrome and 25% with multifocal abnormalities (Confavreux et al 2000). There is no difference in the predictive value of unifocal or multifocal presentation, but there appears a longer period to the second episode with optic neuritis presentation than with either brainstem or spinal cord presentations (Confavreux et al 2003). After the onset of CIS, 70% of patients experience further episodes during which MS symptoms worsen and then symptoms gradually diminish in severity until the next attack. Patients enter the RR phase. These RR episodes typically occur at a random frequency and for an unpredictable period involving the same or different CNS regions. Recovery after these relapses may not be complete. Over time, symptoms can increase in severity and disability becomes more pronounced. This RR phase lasts on average about 20 years, being shorter for males and those older at onset of MS (Tremlett et al 2008). The rate and severity of disease progression during RR phase may differ considerably among patients, with as many as 20–30% continuing to work 20–25 years after disease onset and with only minimal cognitive impairments (Filley et al 1990). With frequent relapses, however, a chronic progressive form of the disease may evolve. The patient is in the SP phase. The age at which individuals enter the SP phase is independent of the initial disease course (Koch et al 2007).

A small proportion of MS patients (about 10%) experience a benign course of MS after CIS; that is, further episodes are delayed for 5–10 years with minimal signs and symptoms in each episode. A benign course of MS is significantly associated with female sex, younger age of onset and absence of motor symptoms at presentations (Costelloe et al 2008). On the other hand, about 10% of patients experience a clinical course called primary progressive (PP) which is characterized by a progressive accumulation of neurological deficits from onset without relapse or remission (Dujmovic et al 2004). This group of patients has a relatively later age of onset and a lesser female preponderance compared with the general MS population.

In population studies, at the one time approximately one-third of individuals are in a quiescent phase of the disease and not significantly disabled, a further third are slowly deteriorating, and the remainder are stable but disabled having had the disease for many years (Swingler & Compston 1992). MS tends to impact on the quality of life rather than on its duration, life expectancy only decreasing slightly in individuals with MS. However, in very disabled individuals, the probability of death is more than four-fold that of the general population.

CLINICAL SYMPTOMATOLOGY

The pattern of symptoms of MS is complex, variable and unpredictable.

Sensorimotor impairments

Weakness in MS may develop gradually in one or more limbs, increasing with use and often described as a feeling of heaviness and clumsiness. Depending on the site of the lesion, signs of an upper motor neuron lesion (Ch. 8) may be present while involvement of the cerebellum and its connections produces ataxic symptoms (Ch. 9), which usually occur in combination with corticospinal damage. Spinal demyelination causes progressive weakness in both legs. In individuals where there is extensive demyelination adjacent to the dorsal root entry zones, lower motor neuron signs may be present.

People with MS who suffer from *loss of muscle activation and control* often have difficulty when participating in activities of daily living and leisure activities. This, in turn, results in a gradual decrease in physical activity. It has been shown that physical limitations are positively correlated with physiological changes in people with MS, which have been shown to be similar to those that occur in healthy people who have experienced prolonged physical inactivity.

Spasticity (along with *fatigue* and *weakness*) is one of the three most common physical signs and symptoms experienced by MS patients. Epidemiology studies show that spasticity is a significant problem for about 60–80% of people with MS (Rizzo et al 2004) and is a major contributor to disability in this population (Beard et al 2003). The most commonly used definition of spasticity is probably that of Lance (1980): 'a motor disorder characterized by a velocity-dependent increase in tonic stretch reflexes (muscle tone) with exaggerated tendon jerks resulting from hyperexcitability of the stretch reflex as one component of upper motor neuron syndrome'. Increasingly, a distinction is being made between resistance to passive movement due to reflex hyperactivity, and resistance resulting from increased mechanical stiffness. The Ashworth Scale (Ashworth 1964) is typically used to measure spasticity, although this scale cannot differentiate between intrinsic muscle stiffness and reflex hyperactivity. As discussed in Chapter 8, increased resistance to passive movement can be caused by an increase in passive stiffness of soft tissue, an increase in stiffness mediated by the stretch reflex, or an increase in intrinsic stiffness which reflects the stiffness of the contractile properties of the engaged crossbridges.

Sinkjaer and colleagues (1993) measured the passive, the intrinsic and the reflex-mediated response to stretch in the ankle extensors and flexors in subjects with MS and able-bodied subjects. The major findings suggested that spastic muscles in individuals with MS have an increased non-reflexive stiffness (passive and intrinsic) and that reflex-mediated stiffness in the ankle extensors during a sustained voluntary contraction does not differ significantly from able-bodied subjects. However, in this study the authors did not directly investigate the passive

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properties of muscles as the measures of stiffness were obtained from electrically stimulated muscles – the muscles were not relaxed. Moreover, the authors examined stiffness at the ankle joint. Ankle stiffness is likely to be due partly to the stiffness of muscles but also to other structures that cross the joint such as ligaments. Recently, Hoang and colleagues (2007) have developed a method that allows direct measurement of passive properties of human relaxed gastrocnemius. Using this method, they showed that the passive properties of gastrocnemius in people with MS (patients are still ambulant and have spasticity) are not different from those in healthy people (Hoang et al 2009).

Altered *sensation* occurs at some stage in almost every individual with MS. Sensory symptoms, such as paraesthesia of a limb or of the face, with numbness, tingling or burning, may be the first clinical signs. Due to the unpredictable nature of the disease, sensory deficits may affect one limb, one side of the body or all four limbs. Involvement of the posterior columns of the spinal cord results in impairments in position and movement sense, vibration sense and touch. Temperature sense may also be affected, resulting in typical heat or cold sensitivity symptoms in MS.

Pain is another common symptom in people with MS. Pain can be the direct result of demyelination and axonal loss (neurogenic pain), or a secondary consequence of another MS symptom (nociceptive pain). Several studies have reported that the incidence of pain within any one month period is between 60–80% of MS patients. Chronic pain, defined as pain of 3 months duration, is experienced by 65–70% of people with MS. Of these, 60% had chronic dysaesthetic pain and 70% experienced episodic pain. Prevalence of pain tends to increase with age and number of years since disease onset. Some studies have noted a higher pain prevalence in females with MS (Hadjimichael et al 2007) and in people who have moderate to severe mobility restriction as measured by the Expanded Disability Status Scale (EDSS) (Ehde et al 2006). However, it is important to note that pain affects people with MS at all stages of the disease, including people newly diagnosed.

Cognitive and affective symptoms

There has been considerable development in the understanding of cognitive impairments in MS over recent years. It is estimated that cognitive impairments occur in up to 65% of MS patients (Winkelmann et al 2007). Cognitive dysfunction strongly affects patients' ability to work, social relationships and quality of life. In a cross-sectional study of 250 MS patients, Lynch and colleagues (2005) found that cognitive impairment is closely associated with physical disability and this relationship appears to be stable throughout the duration of MS.

Results of recent analytical reviews suggest that the effect of MS on cognition is both general, that is all cognitive domains are affected, and specific, that is the effects are larger for the domains of mood, motor functioning, memory and learning (Prakash et al 2008). It was estimated that 40–60% of MS patients suffer deficits in memory and learning (Grafman et al 1990), even at early stages of the disease. Lezak's (1995) clinical observations suggest that slowed mental processing makes it difficult for individuals with attention deficits to grasp all aspects of a verbal message, particularly when long, complicated and delivered rapidly in a complex environment. When patients have no recollection of what has been said or what is happening around them, they and their families interpret this as a problem of memory rather than slowed processing of information. Once patients, families and health professionals understand the nature of the problem, careful attention to how, when and where information and messages are given and activities organized may greatly improve the patient's 'memory'.

Personality and psychosocial behaviour

In addition to cognitive and affective symptoms, MS may be associated with a number of behavioural changes. For example, changes in personality, preferences and attitudes tend to accompany impairments in focusing attention and distractability. Individuals with MS describe feelings of being mentally blocked, of dissatisfaction with themselves and of diminished spontaneity of action (Lezak 1995). Individuals with MS have been described as prone to 'euphoria'. Euphoria may be defined as a fixed state of wellbeing where patients may express a conviction that all is well and that they are physically fit and healthy despite the presence of considerable physical disability. The term euphoria is therefore inappropriate when applied to individuals with MS who attempt to face the future with courage (Compston 1993a). True euphoria is a relatively rare phenomenon, typically associated with advanced disease involving the frontal lobes.

Depression is said to be more commonly observed in patients with MS than in those with comparable medical disorders (Minden & Schiffer 1990). The lifetime risk for depression among MS has been estimated at 50% (Patten & Metz 1997) compared with a lifetime risk in the general population of around 10–15%. Although this is not surprising, given depression is an appropriate reaction to what can be a devastating disease, the appearance of depression does not seem to be related to the severity of the disease. Recent research suggests depression develops after disease onset and is quite stable longitudinally. Because of the stability of depression in MS and the fact that it is unlikely to remit without treatment, it can have devastating long-term consequences for the patient's day-to-day functioning (Arnett et al 2008). Recent work has suggested anxiety disorders are also common in patients with MS, but are frequently overlooked and undertreated.

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Risk factors include being female, a comorbid diagnosis of depression and limited social support (Korostil & Feinstein 2007). Fortunately, anxiety disorders represent a treatable cause of disability in MS.

On the other hand, *stress* may exacerbate symptoms and precipitate onset. There is increasing evidence that stressful life events correlate with exacerbations in MS (Gold et al 2005). A systemic meta-analysis of 14 prospective studies found a consistent association between stressful life events and subsequent exacerbation in MS but there is no link of specific stressors to exacerbations (Mohr et al 2004). From several clinical studies and animal models, it was suggested that stress may be involved with the reduced sensitivity to glucocorticoid and β-adrenergic modulation, which can exacerbate overshooting inflammation in MS. The roles of the two major stress-response systems, the hypothalamic–pituitary–adrenal axis and the autonomic nervous system are topics of interest in current research on stress.

Special senses

The involvement of the visual pathways is very common. The episodic visual blurring so often described by patients early in the disease may later deteriorate further, with some patients losing sight in one eye or suffering double vision. Deafness is more often seen in individuals with established disease. Acute vestibular symptoms with severe positional vertigo (an illusion of movement in the person's relationship with the environment), vomiting, ataxia and headache are typically seen in acute brainstem demyelination. Other senses such as taste and smell may be involved.

Fatigue

Fatigue is among the most common, yet least understood, reported symptoms of MS (Krupp 2003). It may be more disabling than any of the milder symptoms, compromising the person's efficiency and sense of wellbeing (Lezak 1995). Individuals with MS report that fatigue generally occurs daily, interferes with physical and social function and worsens with heat. Individuals with MS, their families and friends can misjudge the impact of fatigue by misinterpreting it as laziness. Individuals who are working complain that they have no energy for recreational activities because of needing to rest at the weekend. Furthermore, fatigue is one of the two major reasons for unemployment among MS individuals (Edgley et al 1991). Recent data from an ongoing longitudinal study (Multiple Sclerosis Society of Australia 1990*), which involves more than 3000 patients across Australia, show that fatigue and impaired mobility are two main reasons for job loss among patients. Many patients are unable to be actively engaged for more than a few hours without fatigue and tend to limit their activity to avoid fatigue and overheating. A vicious cycle occurs, with decreased physical and social activity tending to have a further deleterious effect.

Schapiro and colleagues (1987) have described four types of fatigue: fatigue following *physical exertion*, experienced by the general population and which recovers after a period of rest; *nerve impulse* fatigue following extreme activity which, again, recovers with rest; fatigue related to *depression*, and associated with sleep disturbances, low self-esteem and mood fluctuations; and *lassitude*, or an abnormal sense of tiredness of unknown aetiology. All four types described may contribute to fatigue in MS. However, the sense of tiredness or lassitude is little understood and people with MS appear to be particularly vulnerable to this. However, an additional cause of fatigue is the underlying slowing of nerve impulses along partially demyelinated axons.

Krupp and colleagues (1988) devised the Fatigue Severity Scale (FSS), a nine-item questionnaire in which patients rate their agreement with statements that distinguished fatigue in MS from healthy controls. This scale has been shown to have acceptable internal consistency, stability over time and reflects the effect of fatigue on daily functioning. Interestingly, Krupp and colleagues (1989) found that the severity of fatigue did not correlate significantly with depression in individuals with MS, suggesting that fatigue and depression are separate, although overlapping, entities. Another common scale used to measure fatigue in MS is the Modified Fatigue Impact Scale or MFIS. This is a 21-item scale, developed by US National MS Society, derived from the original 40-item Fatigue Impact Scale. It has been classified as a multidimentional scale and is intended to analyse different aspects of fatigue by assessing impact on physical, cognitive and psychosocial functioning (Tellez et al 2006)

Heat in the form of hot weather, overheated rooms, immersion in hot water and increased core temperature after strenuous physical activities increases fatigue as well as other symptoms of MS, tending to weaken the individual (Lezak 1995). Increased temperature sensitivity, with a reduction in the safety factor for conduction in partially demyelinated axons, may explain the temporary increase in the severity of symptoms experienced by patients after exercise or hot baths (Honan et al 1987). MS patients may experience a temporary increase in the intensity of sensory symptoms immediately post-exercise (Smith et al 2006) but this is unlikely to have any deleterious changes in fatigue and function. On the other hand, cold may help to improve performance. Cooling nerve fibres has been shown in laboratory-based models and in some small studies of people with MS to improve the speed of messages passed along the nerves, and

*http://msaustralia.org.au/msra/research/ms-life-study.php.

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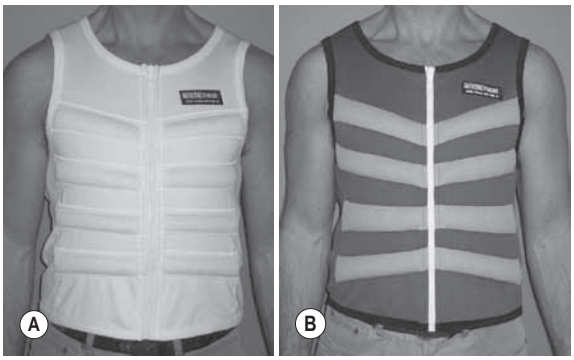

Figure 14.2 A cooling vest is useful for MS patients who suffer heat intolerance.

consequently improves symptoms. There are cooling products available that may assist in decreasing or maintaining a lower body temperature to prevent worsening of symptoms (Fig. 14.2).

Autonomic involvement

Autonomic involvement occurs in most patients with MS. Bladder symptoms are more common in women than men. Impotence may be prevalent in males. Sphincter control may be lost or impaired. Disinhibition causes urgency and frequency which lead to incontinence. Bowel incontinence may also be present. Cardiovascular autonomic dysfunction is usually of minor clinical importance. However, orthostatic intolerance may be present in approximately 50% of patients (Flachenecker 2007) and can easily be detected by routine measurements of heart rate and blood pressure during rest and during standing.

Other manifestations of brainstem involvement

Compston (1993a) reports several manifestations of brainstem demyelination. Abnormalities of eye movement are common in MS. Facial palsy is reported to be present in individuals with established MS. Extensive brainstem demyelination may produce disturbances of consciousness or central respiratory failure. Paroxysmal episodes (tonic spasms), in which painful tetanic posturing of the limbs occurs for up to 1 or 2 minutes, may also be present. In a study on 483 patients, Zadro and colleagues (2008) found that isolated cranial nerve involvement was present in 10% of patients. The trigeminal nerve was most frequently involved, followed by facial, abducens, oculomotor and cochlear nerves. However, only 54% of patients had a brainstem magnetic resonance imaging (MRI) lesion that could explain the symptoms.

MEDICAL MANAGEMENT

There is at present no single accurate laboratory-supported diagnostic test for MS. Rather, diagnosis of MS requires use of both clinical and paraclinical criteria. The latter involves information obtained from MRI, motor-evoked potentials (MEPs) and cerebrospinal fluid (CSF) analysis. Investigations such as MRI, CSF analysis in individuals with MS are used to document the site and extent of lesions, to confirm the presence of intrathecal inflammation and to exclude conditions that may mimic demyelinating disease. MRI is thought to have increased the accuracy in diagnosing MS from 60% to 90% (Ebers 1994) and appears to be a more sensitive indicator of disease activity than the clinical history or neurological examination (Polman & Hartung 1995).

New diagnostic criteria for MS integrating MRI assessment with clinical and other paraclinical methods (the McDonald criteria) were introduced in 2001 (McDonald et al 2001). Since then, these criteria have been extensively assessed and used. The 2005 revisions to the McDonald diagnostic criteria for MS aim to simplify and speed diagnosis while maintaining adequate sensitivity and specificity, and are now widely accepted by the neurology community.

Medical treatments in MS can be divided into several categories. The first category is the use of medications that impact the underlying disease (disease-modifying therapies), targeting some aspect of the inflammatory process of MS with an aim of preventing inflammation which causes relapses. The second one is medications that help to decrease the severity and duration of MS relapses (steroids), which aim to suppress inflammation. The third category is medications that help ease many MS-related symptoms such as fatigue, spasticity and pain, to name a few.

A large body of experimental evidence implicating immune-mediated processes in activation and progression of MS has led to a search for immunotherapies that not only abrogate acute relapses, but also modify the progression of the disease. Consequently, many disease-modifying drugs (DMDs) or immunomodulatory treatments for MS have been developed and clinically trialed on a large scale. To date, there are enough data to support widely used DMDs that may have an effect on slowing the accumulation of disability over time. Some of these drugs are interferon beta-1a (Avonex® or Rebif®), interferon beta-1b (Betaseron®), glatiramer acetate (Copaxone®) and the latest natalizumab (Tysabri®). Each of these drugs acts to block different inflammation pathways and head-to-head studies have shown that none of them is more effective than the others. However, treatment may not be an option for every person with MS. DMDs are most effective in people with the RR type of MS while there is no treatment for people with PP MS. In

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addition, resistance to a drug is another concern over a long period of use of DMDs.

A short course of intravenous methylprednisolone is often used for the treatment of relapses in multiple sclerosis. But there is still no evidence to indicate that this influences the eventual outcome of the disease, although it may accelerate the onset of a remission when given during an acute relapse. In many cases, after a course of intravenous methylprednisolone, oral prednisone is prescribed and gradually tapered off. Recent observations suggest that oral prednisone following treatment with intravenous methylprednisolone for an MS relapse does not lead to improved neurologic outcome after 12 months compared with treatment with intravenous methylprednisolone only (Perumal et al 2008).

Drugs such as baclofen and diazepam may be useful in the reduction of hyperactive reflexes (spasticity). When given to individuals experiencing severe flexor and adductor spasms associated with spinal cord lesions, these agents may reduce hyperactivity sufficiently to enable the individual to sit and stand more comfortably or to be nursed more easily. However, orally administered drugs such as these, especially at high dose, can cause severe side-effects such as general muscle weakness. Studies using botulinum toxin to relieve adductor spasms have reported spasticity relief in the lower extremities and significantly increased range of passive movements in joints, making nursing care and rehabilitation easier with no significant side-effects from the toxin (e.g., Snow et al 1990; Sobolewski 2007).

Documenting the course, assessing the prognosis and validating the effects of intervention in MS require standardized measures of disability. Until recently, the most extensively used scale appears to be the Disability Status Scale (Kurtzke 1983). However, this scale is widely criticized on theoretical grounds, on its insensitivity to change, its emphasis on ambulation while neglecting other relevant functions and on its lack of objectivity and reliability (e.g., Goodkin et al 1988; Polman & Hartung 1995; Kragt et al 2008). Wade (1992) suggests that one should only consider using the Kurtzke scale in order to be able to compare results with other studies. He further suggests that it is preferable to measure change in specific impairments and motor performance using measures that are valid and reliable (see Ch. 3).

PHYSIOTHERAPY: AN OVERALL VIEW

Many of the challenges associated with progressive neurological disease are well exemplified in MS. Progressive disease presents a series of difficulties and burdens that impact on the quality of life of the individual, their families and health professionals. Coping with functional limitations, explicit uncertainty about the future and possibility of loss of function and family role has a negative impact on quality of life, social interaction, vocational and leisure activities and is extremely stressful.

Almost any manifestation of neurological impairment can be observed in individuals with MS (see Chs 8–10). The nature of impairments and deficits in functional motor performance that arise from one individual to another, but also in the same individual at different times, are wide ranging, variable and unpredictable. Physiotherapists. as part of a team working with individuals with MS, need to be sensitive to the individual's wishes in working out strategies to enable that individual to preserve dignity and lifestyle. The physiotherapist needs to have sufficient knowledge and expertise to address wide-ranging problems, and the flexibility to cope with the unpredictable nature of the disease. If physiotherapy is to be effective, it needs to be focused and demonstrate clearly to the patient the benefits of interventions.

The overall aims of physiotherapy as part of a multidisciplinary team are to:

- optimize performance in everyday activities and skills
- maximize functional ability
- prevent unnecessary disability and handicap (Fig. 14.3C–E)
- improve the individual's quality of life.

The reader should refer to Chapters 2 and 4–7 for task-related training.

More specific aims for an individual may be to:

- preserve/improve muscle strength
- preserve/improve aerobic capacity
- preserve musculoskeletal integrity
- manage fatigue
- collaborate with the individual to ensure that interventions are relevant to the person's needs and desires
- collaborate with the individual in setting goals and defining expectations
- provide necessary aids to living.

Strengthening exercise

The need to prevent secondary adaptation to disuse is critical for the maintenance of performance of everyday actions as well as for continued participation in social and vocational activities. There has been no intervention that has proven effective in modifying long-term prognosis in multiple sclerosis but exercises are now accepted to be an important part of symptomatic treatments for people with MS (Rietberg et al 2005). Until about 20 years ago people with MS were often advised to avoid strenuous physical activities or exercises on the basis that they could worsen signs and symptoms of MS and even increase the disease activities. However, in the past 20 years a growing body of evidence has indicated that exercises are beneficial for

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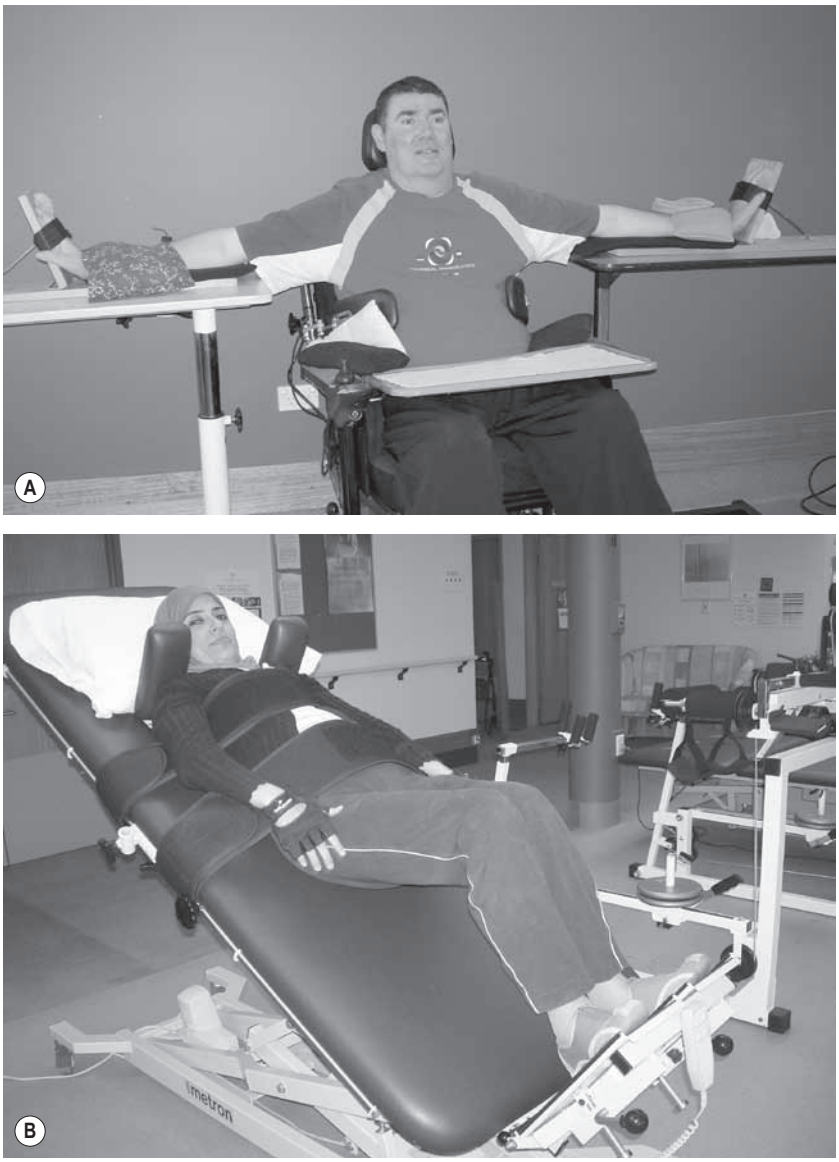

Figure 14.3 (a) Positioning can be used to prevent contracture in people with advanced MS; (b) A tilt table can be used to strengthen knee extensors if the patient is too weak to practice standing up from a seat. It can help prevent/slow down ankle contracture;

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Figure 14.3 Continued (c) Specific equipment is used for upper limb strengthening exercise for people in a wheel chair; (d) Weight is used to help reduce tremor and ataxia during functional exercise of the upper limb;

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Figure 14.3 Continued (e) Eccentric exercise by walking backward and downhill on a treadmill can help improve dorsiflexion and passively stretches the calf muscle in people with MS who are still walking.

people with MS not only at impairment level but also at functional and participation levels. Strength and cardiorespiratory exercises are often prescribed to people with MS who have difficulties performing activities of daily living and/or participating in leisure activities. Results from many studies indicate that these difficulties are partly due to the primary effects of MS, such as loss of muscle activation and muscle control, fatigue and spasticity, and partly due to secondary effects, such as from prolonged disuse which results in deconditioning of the muscles and reduced fitness.

There has been considerable historical debate regarding the safety and appropriateness of prescribing strenuous physical exercise for people with MS. Concern has focused on the possibility of exacerbating MS symptoms such as fatigue or weakness which could be counterproductive to day-to-day symptom management. However, there is strong evidence indicating that exercise does not cause prolonged or permanent worsening of MS symptoms (Smith et al 2006), but rather that exercise often results in improvements in a range of MS symptoms (Gallien et al 2007). No adverse effects of strength exercise in the MS population have been documented in the literature.

There is strong evidence showing that strength training in people with MS can improve muscle force production (DeBolt & McCubbin 2004; Taylor et al 2006). Other benefits of strengthening exercise are improved walking speed and endurance (Taylor et al 2006), improved gait kinematics, reduced fatigue and physical and social disability (White et al 2004).

Some people with MS have reported that exercise preceded an MS exacerbation. However, no studies have been able to identify exercise as the cause of an exacerbation. Exercise can result in a temporary increase in existing symptoms or onset of new previously silent symptoms in people with MS (Smith et al 2006). This is probably related to a heat-induced reduction in nerve conduction velocity. Such symptoms tend to resolve within 30 minutes of rest.

Aerobic exercise

The benefits of regular exercise in the able-bodied and disabled population are well documented and include: increased wellbeing and improved mental state; increased cardiorespiratory fitness; reduction in depression and anxiety: and decreased excess body fat.

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In the past two decades a large body of research has investigated the benefits of cardiorespiratory, or endurance, exercise in people with MS. In a recent review by Dalgas et al (2008), at least 14 studies that have evaluated the effects of endurance training in people with MS were identified. Although there is general methodological weakness in these studies, some important clinical findings can be derived. Endurance exercise at low to moderate intensity is well tolerated and has potential physiological and psychological benefits among people with MS. However, these studies have primarily involved subjects who are minimally to moderately physically disabled with an expanded disability status scale (EDSS) score of less than 7. A single case study investigated the response to arm crank aerobic training in a wheelchair participant (EDSS 7.5) (Smith & Hale 2007). Although the subject demonstrated a trend toward improving cardiorespiratory fitness after 3 weeks of exercise, the study had to be terminated early due to the subject developing shoulder pain.

On the other hand, concerns have been voiced relating to the safe undertaking of cardiorespiratory testing and fitness training in older and long-term sedentary groups who have a number of other risk factors. In recognition of these issues, the American College of Sports Medicine (2006) recommends allocating exercise participants into one of two streams for cardiorespiratory testing and exercise prescription:

- High risk: Individuals with more than two risk factors for cardiorespiratory disease need formal testing to ascertain actual working heart rate to achieve VO_2 maximum. This eliminates the safety issues relating to blunted heart rate response, poor systolic elevation and cardiac risk factors. Exercise prescription is based on the measures obtained through this testing.
- Low to moderate risk: Individuals with two or less risk factors for cardiorespiratory disease can be prescribed exercises using heart rate and supplementary exercise intensity measures such as the Borg Rate of Perceived Exertion (RPE) (Borg 1970) scale and blood pressure monitoring.

Clinicians should adopt these guidelines when testing and prescribing exercise for people with MS who are inactive. Individuals with high risk should have a medical clearance from their treating doctors and should start at low intensity.

Many benefits from cardiorespiratory training have been identified in people with MS. A particularly interesting pioneer paper on this topic (Petajan et al 1996) demonstrated a positive impact on cardiovascular responses and facets of wellbeing in individuals with MS. The results indicated significant improvement in cardiovascular fitness, with increases in VO_{2max} comparable with those reported in studies involving healthy sedentary subjects. The improvement did not seem to be related to the degree of neurological impairment. Although training did not involve *specific* strengthening exercises, increases in strength of most upper limb muscles and knee extensors were reported, indicating that muscle strength can be increased in a demyelinating disease. Similar results were also reported by Mostert & Kesselring (2002). Other benefits from aerobic exercise include improved lung function, improved activity levels and aerobic thresholds (Mostert & Kesselring 2002).

To date, no studies have reported adverse cardiac symptoms in response to cardiorespiratory testing or exercise participation in people with MS. This may be due to the testing of predominantly minimally to moderately physically disabled individuals (EDSS 0–6), or because most studies commenced cardiorespiratory training within the 50–60% of maximal predicted heart rate range with gradual increases in intensity over the training period.

Some particular symptoms that may prevent people with MS from undertaking aerobic exercise include fatigue, heat intolerance and spasticity. These symptoms can be managed to a certain extent for optimal patient compliance and adherence to exercise.

Preserving musculoskeletal integrity

Individuals need advice and instruction about what muscles and soft tissues are at risk of shortening and becoming stiffer, what stretches to do and how often. In the very disabled person, this may require equipment (see Fig. 14.3A, B and E). It is interesting to note that subjects participating in a baclofen drug trial reported that muscle stretching provided them with a subjective sense of control over stiffness (Brar et al 1991), although the stretches were performed for only a short period which would not be considered to be of sufficient duration to prevent contracture in inactive individuals (Williams 1990). People who are wheelchair bound should spend at least 30 minutes a day stretching hip and knee flexors and ankle plantarflexors. A tilt table is probably essential for some individuals in order to prevent severe contractures (see Fig. 14.3B).

Managing fatigue

Management of fatigue involves discussion with the individual and gathering of information about the nature, extent and precipitating factors in fatigue. Thus, managing fatigue in MS requires a multidisciplinary approach. In general, strategies to implement may include energy conservation, exercise, equipment and environmental modifications.

To conserve energy, people with MS, their families and carers should be educated about balancing rest and activity through prioritizing, delegating, pacing tasks and taking regular short rests between activities. The patient can learn how to identify aggravating lifestyle factors and apply the above strategies as their activities change and as

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the effects of fatigue vary over time. Practical strategies for providing exercise and treatment sessions when the person with MS is affected by fatigue include scheduling the exercise or treatment sessions early in the day, encouraging the client to work at their own pace and have regular rests, beginning with a short duration (<20 minutes) and gradually increase as tolerated. Similarly, beginning exercise with a low number of repetitions and low intensity then gradually progressing as the participant can manage more repetitions or increased intensity, mixing vigorous exercises with lighter exercises to avoid long periods of exercise at high heart rates, alternating muscle groups to rest one muscle group while exercising another and keeping body temperature down as much as possible can help. Assistive equipment for mobility, self-care and seating should be used to minimize energy consumption during activities of daily living. Changes to environmental factors, for example installing air conditioning, lifts (elevators), railings in stairwell and ergonomic office equipment, can also assist in minimizing fatigue.

There are some pharmacological agents that may help reduce fatigue in MS. However, these medications are not widely used in Australia for MS fatigue management compared with other parts of the world.

Coping with reduced activity and participation

Limitations in community participation for people with MS, including the extent of environmental modification, facilities for ongoing exercise, social, vocational and recreational availability, are to some extent dependent on national health policy. Health professionals have an important role in being familiar with and referring patients to the appropriate agency, which may include: a MS society; an independent living centre; technical aids to the disabled: accessible leisure centres; respite care provision: sporting groups for the disabled; self-help groups; carer support groups; and vocational rehabilitation services. These agencies can be valuable sources of information.

Health professionals also need to keep up to date with technological advances designed to increase environmental interaction to improve the quality of life for individuals with MS and their families. Some recent publications are helpful; for example, The Multiple Sclerosis Society of Australia has published a booklet (1990) for the newly diagnosed person or a series of handouts to help health professionals better understand specific problems in MS and provide guidelines for management of these problems.

The stress on families assisting an individual with MS can be high. A survey in the UK, for example, found that 15% of carers were depressed and 24% suffered from clinical anxiety (University of Southampton 1989). Information regarding availability of services, practical help at home and access to respite facilities appear to be some of the more important factors to address in order to relieve family stress. A hoist (Fig. 14.4) enables the person to be moved, thus diminishing some of the physical stress on family members.

A wheelchair is necessary for individuals who may be coping with walking about the house but who otherwise are confined to the house. Wheelchair training should be instituted to train the individual to negotiate a more complex environment than the home. An electric wheelchair and/or a three-wheel scooter give more freedom to a person who is unable to walk effectively and efficiently or is having difficulty operating a hand-driven chair. In some instances the energy required to walk may be better put to more beneficial activities, and the use of a wheelchair or scooter may be critical to remaining employed and maintaining social contact outside the house.

Measurement

Measures of motor performance need to be reliable and valid and be sensitive to changes important for the individual (Ch. 3). Paltamaa and colleagues (2008) found that

Figure 14.4 A hoist decreases the physical burden for carers and ensures greater comfort for the individual when being moved from one place to another.

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the measures most responsive to deterioration in MS were self-reported scores in self-care, mobility and domestic life; distance walked and change in heart rate during a 6-minute walk test; 10-m walk test speeds, stride length and cadence; repetitive squatting; and box-and-block test scores. MS-specific measures have been developed in recent years. One is the Multiple Sclerosis Spasticity Scale (MSSS-88) that is used to measure spasticity from a patient's perspective (Hobart et al 2006). The Modified Fatigue Impact Scale (MFIS) is a modified form of the Fatigue Impact Scale (Fisk et al 1994) based on items derived from interviews with MS patients concerning how fatigue impacts their lives. This instrument provides an assessment of the effects of fatigue in terms of physical, cognitive and psychosocial functioning. The full-length MFIS consists of 21 items while the abbreviated version has five items. The abbreviated version can be used if time is limited but the full-length version has the advantage of generating subscales. Although these measures have been shown to be valid and clinically relevant, further research is needed to establish their reliability.

Coping with chronic illness and uncertainty can be very stressful. Schwartz and colleagues (1996) have developed and tested a *self-efficacy scale* specifically for MS. Self-efficacy is a term used to describe the degree of confidence an individual has in his/her ability to cope with a specific situation or condition. The test was found to have high internal consistency and test–retest reliability. This scale should be valuable in measuring the outcome of intervention in terms of enhancing the patient's wellbeing and confidence. High self-efficacy in individuals with MS has previously been found to be associated with better adjustment and less psychological dysfunction (e.g., Schnek et al 1995).

The use of *integrated care pathways* (ICPs) appears to be useful in identifying key factors influencing outcome of intervention. An ICP is a document that details the interventions expected to occur during a specific period of clinical intervention (Rossiter & Thompson 1995). Several recent audits have shown that the introduction of ICPs in an inpatient setting resulted in decreased length of stay, a framework for identifying not just how many goals were achieved but also for identifying why goals were not met, and improved record keeping (Rossiter & Thompson 1996).

In summary, as Compston (1990) has pointed out, a clearer understanding of the epidemiological and biological factors that determine the geographical and anatomical dissemination of MS may provide a solution to this disease. The most exciting development in neurological rehabilitation for persons with MS is the impact of strengthening and aerobic training, with adequate temperature control, on strength and fitness, together with factors related to quality of life. Therapists are encouraged to become familiar with the literature on strengthening and aerobic training in order to prescribe effective exercise programmes. Other critical interventions include prevention of unnecessary limitations in activity and participation and advice on reducing unnecessary fatigue in everyday, social and vocational activities.

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